

CS6360-Assignment 7

Theorem and Proof - Port Hamiltonian Neural Networks (PHNN)

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Statement

A model that captures real world dynamical systems composed of many subsystems requires a general compositional learning framework which can be based on energy flow dynamics. So far Port Hamiltonian (PH) based neural networks are proposed to learn such dynamical systems from data. Though they work well with dissipative systems. Here, we propose an alternative incremental Port Hamiltonian Neural Networks (iPHNN) to deal with conservative systems. First, try to replace PH by maximal monotone relation and relate it with continuous time gradient descent for efficient optimizations methods. Second, we propose to use Stormer–Verlet port Hamiltonian gradient flow (SV-PHGF) based learning optimizer which is computationally more efficient than Adam and RAdam.

Dirac Structure

Definition

A (constant) **Dirac structure** is a subspace

$$\mathcal{D} \subset \mathcal{F} \times \mathcal{E},$$

where $\mathcal{F}$ is a linear space with $\mathcal{E}$ its dual where $\mathcal{E} = \mathcal{F}^*$

such that

- 1) $\langle e|f \rangle \geq 0$ for all $(e,f) \in \mathcal{D}$. where $\langle e|f \rangle = e^T f$
- 2) $\dim \mathcal{D} = \dim \mathcal{F}$.

Monotone

Definition

A relation $\mathcal{M} \subseteq \mathcal{F} \times \mathcal{E}$ with $\mathcal{E} = \mathcal{F}^*$ is said to be

- *monotone* if

$$\langle e_1 - e_2 | f_1 - f_2 \rangle \geq 0$$

for all $(f_i, e_i) \in \mathcal{M}$ with $i \in \{1, 2\}$

- *cyclically monotone* if

$$\langle e_0 | f_0 - f_1 \rangle + \langle e_1 | f_1 - f_2 \rangle + \dots + \langle e_m | f_m - f_0 \rangle \geq 0$$

for all $m \geq 1$ and $(f_i, e_i) \in \mathcal{M}$ with $i = 0, 1, \dots, m$

- *maximal monotone* if it is (cyclically) monotone and $\mathcal{M}'$ is monotone and $\mathcal{M} \subseteq \mathcal{M}' \Rightarrow \mathcal{M} = \mathcal{M}'$ holds.

Introduction

Main Idea: Replace the combination of a **Dirac structure** and **dissipation** by a maximal (cyclically) monotone relation.

Definition

Consider a linear space $\mathcal{F}$ with $\mathcal{E}$ its dual, and a linear output space $\mathcal{Y}$ with dual space $\mathcal{U}$. Consider a maximal monotone relation

$$\mathcal{M} \subset \mathcal{F} \times \mathcal{Y} \times \mathcal{E} \times \mathcal{U}$$

and a Hamiltonian $H: \mathcal{F} \rightarrow \mathbb{R}$. Then the dynamics of the corresponding **(maximal) cyclically monotone port-Hamiltonian system** is given by

$$(-\dot{x}(t), y(t), \nabla H(x(t)), u(t)) \in \mathcal{M}, \quad t \in \mathbb{R}$$

where $\nabla H(x) := \frac{\partial H}{\partial x}(x)$.

Theorem

Statement : A monotone port Hamiltonian system defined over an ordinary and differential convex function gives rise to a non linear monotone port Hamiltonian dynamical system composed of a convex function and Hamiltonian that can be used to compose a dynamical system consisting of conservative sub systems and trained using port Hamiltonian gradient flow methods.

Proof

If $\mathcal{M}$ is maximal **cyclically** monotone then it gives rise to maximal cyclically monotone pH systems

$$\begin{aligned}\dot{x} &\in -\partial_e K(\nabla H(x), u) \\ y &\in \partial_u K(\nabla H(x), u)\end{aligned}$$

for some convex function $K: \mathcal{E} \times \mathcal{U} \rightarrow (-\infty, +\infty]$.

and if K is an ordinary and differentiable function, then we get a nonlinear class of systems with convex function K and Hamiltonian H .

$$\begin{aligned}\dot{x} &= -\frac{\partial K}{\partial x}(\nabla H(x), u) \\ y &= \frac{\partial K}{\partial u}(\nabla H(x), u)\end{aligned}$$

Proof

Consider the problem of minimizing a convex function $P: \mathbb{R}^n \rightarrow \mathbb{R}$.

The, gradient algorithm in **continuous time** is given as

$$\begin{aligned}\tau \dot{q} &= -\frac{\partial P}{\partial q}(q) - Bu \\ y &= B^T q\end{aligned}$$

where τ is a positive definite matrix determines the time-scales of algorithm.

This defines a **maximal cyclical monotone** pH system

$$\dot{x} = \frac{\partial P}{\partial q}(q), \quad q = \nabla H(x) = \tau^{-1}x$$

with quadratic Hamiltonian $H(x) = \frac{1}{2}x^T \tau^{-1}x$

q : state, B : $n \times m$ matrix, $u \in \mathbb{R}^m$: input vector, output vector as $y = B^T q \in \mathbb{R}^n$.

Proof

Extending to include input and outputs by considering

$$K(q, u) = P(q) - q^T B u + \frac{1}{2} u^T D u$$

which is convex in q, u if P is strongly convex and D is large enough. Resulting **maximum cyclically monotone pH system** is

$$\begin{aligned} \dot{x} &= -\frac{\partial P}{\partial q}(q) + B u, & q &= \tau^{-1} x = \nabla H(x) \\ y &= -B^T q + D u \end{aligned}$$

can be used for **distributed** optimization.

Proof

In case if $D=0$, maximal monotone (not cyclically) pH system

$$\begin{aligned}\dot{x} &= -\frac{\partial P}{\partial q}(q) + Bu, & q &= \tau^{-1}x \\ y &= B^T q\end{aligned}$$

Where, K is a convex function of P , and both $\dot{x}, y$ are functions of (q, u) . A **monotone pH system** composed of a convex function K and Hamiltonian function H is given by

$$\begin{aligned}\dot{x} &= -\frac{\partial K}{\partial x}(\nabla H(x), u) \\ y &= \frac{\partial K}{\partial u}(\nabla H(x), u)\end{aligned}$$

Conclusion

- Replaced the Dirac and dissipation relation by maximal monotone.
- System transition from pH system to incremental pH system.
- Stated a theorem based on differentiable convex function.
- Maximal cyclically monotone relation gives a monotone pH system consisting of a convex function along with a Hamiltonian system.
- Composition of two monotone relation with partially shared variables is again a monotone which can be used to build composite systems.
- An incremental pHNN can be defined using monotone pH system.